

AccountingBench: A Structured Benchmark for Systematic Evaluation of Large Language Models in Accounting Education and Professional Tasks

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Abstract

This paper introduces AccountingBench, a structured and instrumented benchmarking framework for evaluating large language models on accounting tasks in educational and professional contexts. Accounting problem solving is shaped by normative constraints and judgment under uncertainty, which makes reliability and auditability central requirements for responsible use. AccountingBench operationalizes accounting competence through a task taxonomy spanning selected-response and generative formats. As an initial empirical illustration, we report results on 520 validated tasks across financial accounting, managerial accounting, and taxation, benchmarking ten models from five providers (gpt-5.4, gpt-5.2, claude-opus-4-6, claude-sonnet-4-6, gpt-5-mini, Mistral-Large-3, grok-4-fast, gpt-4o, DeepSeek-V3.2-2, and mercury-2) under a harmonized inference and scoring setup. Average performance separates systematically across models, with gpt-5.4 achieving the highest overall score (72.2%), and the largest inter-model gaps observed in tax tasks and mixed regulatory contexts. Beyond aggregate scores, we report confidence distributions and pipeline design choices that enable stability and calibration analyses. The dataset and codebase are available upon request and will be released in a versioned form to support replication and extension.

JEL Classification: M41, O31

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1 Introduction

The growing use of generative AI in accounting has drawn increasing attention to how language models handle complex, domain-specific tasks. Accounting involves structured standards, judgment-based assessments, and compliance-related responsibilities, all of which make the reliability and traceability of AI-supported decisions especially important. As AI tools begin to support tasks such as financial statement analysis, audit procedures, or the interpretation of accounting standards as well as the answering of questions in the field of accounting education, the question arises of how their outputs can be assessed in a consistent and meaningful way.

The increasing importance of AI in specific fields of application raises questions about the validity, traceability, and evaluation of language models (Hutchinson et al., 2022). Particularly in areas where generative AI offers a wide range of applications, consideration must be given to how the tasks assigned to LLMs can be reviewed and evaluated, which raises the need for benchmarking systems (Bean et al., 2025; Bubeck et al., 2023; Hernández-Orallo, 2017). Benchmarking requires well designed use cases that allow stakeholders to evaluate model performance and behavior under controlled conditions. This includes accuracy, reasoning quality, and the degree to which model responses approximate expert judgments (Raji et al., 2021). Because general-purpose benchmarks often fail to capture domain-specific reasoning and may overlook deeper analytical capabilities (Gevers et al., 2025; Guha et al., 2024; Reuel et al., 2024), task-specific benchmarks are increasingly needed.

Accounting benchmarking is often framed as building yet another benchmark (Johanson et al., 2022). We instead treat the benchmark as a measurement instrument for answering accounting-specific research questions about model reliability in normative, compliance-driven settings. Existing finance and accounting benchmarks and exam-based evaluations demonstrate meaningful progress in task coverage and workflow realism, yet they typically foreground aggregate task success and often abstract from three properties that are central to accounting reliance: whether models exhibit hallucinated compliance-confident, standard-like treatments that are substantively wrong, whether outputs are stable and reproducible under repeated inference, which is critical for auditability and educational assessment, and whether exposure to authoritative guidance improves substantive correctness rather than merely increasing linguistic plausibility. Our benchmarking framework therefore operationalizes accounting problem-solving competence as a combination of normative correctness, validity, and justification traceability, and it couples this construct with an execution pipeline that logs confidence and run-to-run variation. In doing so, the benchmark becomes enabling infrastructure for theory relevant evidence on where LLMs can be safely automated versus where

they should be used only for augmentation with human oversight. Early literature on AI in the accounting field set up a human benchmarking loop specifically for ChatGPT and compared the answers of the test-takers with ChatGPT, which provides a preliminary form of benchmarking (Wood et al., 2023). Yet accounting tasks require not only factual correctness but also justified reasoning paths and transparent logic. This highlights the need for large, well-structured datasets that reflect real accounting task solving complexity. In this context, the structure and design of those questions are highly relevant. While multiple-choice or single-choice formats do not necessarily require justification or logical argumentation, calculations, journal entries, and open-ended questions depend on reasoning that makes the solution path, the underlying idea, and the final answer transparent. Insights from legal education already show that closed-form questions often fail to capture deeper analytical thinking or domain-specific reasoning. Multiple-choice formats tend to measure recognition rather than argumentation, so that essays and open formats are better suited to assess how students identify relevant issues, structure their reasoning, and justify their conclusions (Bozin et al., 2021; Brennan, 2001).

Recent studies in AI benchmarking emphasize that benchmarks relying heavily on simplistic or closed-form tasks risk overlooking the actual reasoning capabilities of language models and may encourage superficial performance gains (Gevers et al., 2025; Reuel et al., 2024). Other findings show that many existing benchmarks lack validity because they fail to reflect the complexity of real-world tasks (Eriksson et al., 2025). Work on educational benchmarking demonstrates that diverse and open-ended task formats are necessary to evaluate genuine competence rather than pattern matching. These insights suggest that accounting benchmarks must incorporate more than closed-form or recognition-based items if they are to meaningfully assess whether models can understand accounting logic, interpret standards, and state justified conclusions (Xu et al., 2025).

This paper makes three contributions to the emerging field of domain-specific LLM benchmarking in accounting:

1. It develops a construct-valid operationalization of accounting problem-solving competence tailored to the cognitive and normative requirements of the discipline. The framework explicitly models the components of accounting reasoning and translates them into task types that allow systematic measurement of model capabilities.
2. It proposes a transparent data-generation and evaluation architecture that integrates single- and multiple-choice and open-ended tasks, controlled prompt templates, and scoring procedures, expanding the limitation to closed-form ques-

tions, by emphasizing interpretive reasoning and justification, thereby more accurately reflecting real-world accounting judgment.

3. It establishes a methodology for dataset provenance and difficulty stratification based on representative accounting materials, including textbooks, exam items, and expert-reviewed cases.

The benchmark is explicitly scoped to textual reasoning tasks under IFRS, local GAAP, and local tax law as well as calculations and journal entries. It is designed for research and educational evaluation, not yet for regulatory or audit assurance purposes.

2 Related Work

Traditional benchmarking of LLM outputs, defined as benchmarking by human beings, is not only time-consuming but also resource- and cost-intensive, although manual inspection via error analysis enables engineers to gain rich intuition regarding model behavior and reasoning (Ouyang et al., 2022). Other comparable to accounting domain-specific areas of application include law, finance, and medicine (Chen et al., 2021; Guha et al., 2023; Q. Jin et al., 2019; Pal et al., 2022; Zhu et al., 2021). Benchmarking is typically structured in such a way that, in order to evaluate the performance of a model, the benchmark contains a data set consisting of pairs of model inputs and model outputs (Bean et al., 2025). Model inputs consist of the data the model receives, and model outputs are the ideal correct answers. One of the most comprehensive critiques of modern AI benchmarking practice is particularly the tendency of widely used benchmarks to present themselves as evaluation of general intelligence despite being grounded in narrowly scoped, context-specific datasets (Raji et al., 2021). Here, the central argument is that most benchmarks function as “Everything in the Whole Wide World” (Raji et al., 2021) tests without clearly articulating what phenomena they actually measure. Rather than claiming to measure general reasoning or domain-agnostic intelligence, the accounting benchmark framework is positioned as a context specialized benchmark targeting a specific, professionally defined domain for accounting reasoning. To gain better insights into the structure and frameworks of other domain-specific and -agnostic benchmarks, we compared the main tasks and objectives of established AI benchmarks.

Table 1: Comparison of benchmarks: a selection of AI benchmark frameworks

Name	Publishing Date	Description	Reference
GDPval	2025-10-05	Benchmark on real-world economically valuable tasks across 44 occupations in 9 sectors.	Patwardhan et al. (2025)
HELM	2022-11-16	Domain-agnostic benchmark for holistic and multi-metric evaluation of language models across scenarios and metrics.	Bommasani et al. (2023)
GSM8K	2021-10-27	Domain-specific benchmark for grade-school mathematical reasoning tasks.	Cobbe et al. (2021)
GLUE	2018-04-20	Multi-task benchmark and analysis platform for natural language understanding; later extended by SuperGLUE.	A. Wang et al. (2018)
MedMCQA	2022-03-27	Large-scale multi-subject medical multiple-choice benchmark based on entrance examinations.	Pal et al. (2022)
TAT-QA	2021-05-17	Finance benchmark for question answering over hybrid tabular and textual content with numerical reasoning.	Zhu et al. (2021)
LegalBench	2023-08-20	Benchmark for measuring legal reasoning capabilities in large language models.	Guha et al. (2023)
PubMedQA	2019-09-13	Biomedical question-answering benchmark based on research questions.	Q. Jin et al. (2019)
MedQA	2021	Medical multiple-choice benchmark built from professional medical board examinations.	D. Jin et al. (2021)
BigLawBench	2024-08-29	Benchmark for evaluating large language models on complex, real-world legal tasks.	Harvey AI (2024)

Name	Publishing Date	Description	Reference
BIG-bench	2022-06-09	Large collaborative benchmark probing language-model capabilities across diverse expert-designed tasks.	Srivastava et al. (2022)
MMLU	2020-09-07	Massive multitask multiple-choice benchmark spanning 57 subjects from school to expert level.	Hendrycks et al. (2020)
FinQA	2021-09-01	Benchmark for numerical reasoning over financial data and financial reports.	Chen et al. (2021)

A key reference point for the development of a domain-specific accounting benchmark is the existing literature on **text analysis in accounting**. The literature shows that language-based analyses have been an established research method in accounting and that methodological findings from this field are relevant for the design of a benchmarking framework. According to a survey (Loughran & McDonald, 2016), which highlights the challenges involved in computer assisted analysis of financial texts and the errors that arise when non specialized tools are **applied to specific issues**. It shows that common sentiment or linguistic methods often lead to misclassifications because financial texts have special linguistic patterns, regulatory terminology, and context-dependent meanings (Loughran & McDonald, 2016). This illustrates that the accounting context is not a neutral linguistic space, but is shaped by rules, norms, and specific technical terms. Furthermore, it underscores the need to design evaluation systems that explicitly take these peculiarities into account. If the technical context is ignored in benchmarking, distortions arise because models reproduce general patterns without demonstrating a specific understanding of accounting. Modern NLP and machine learning methods significantly expand the possibilities of accounting research, but at the same time, new problems arise, particularly with regard to validity, interpretability, and methodological consistency. The increasing complexity of models is leading to a renewed focus on **construct-valid design**. The chosen measurement method must measure what it claims to measure. For an LLM accounting benchmark, this means that tasks, scoring protocols, and data **sources must** be explicitly aligned with the cognitive and normative **requirements of accounting** (Bochkay et al., 2023). It is clear that a benchmarking framework should not be primarily driven by technology but must be developed consistently from the logic of **accounting research**.

Recent evidence from finance and accounting tasks underscores that LLM outputs can be materially variable across repeated runs, raising concerns for both scientific

reproducibility and practical deployment. Jiang et al. (2025) study consistency and reproducibility across diverse finance and accounting tasks and highlight that evaluation must account for stochasticity and prompt sensitivity rather than treating a **single run as representative**. This directly motivates benchmarking frameworks that preserve full model call logs, support repeated sampling under controlled settings, and include reliability metrics (e.g. variance, disagreement rates, or stability scores) alongside accuracy. Taken together, the methodological literature implies that an accounting benchmark’s contribution is strongest when framed not only as a new task collection, but as a measurement instrument that supports reliable inference about model capability, error modes, and the causal role of design choices.

A growing body of work already proposes domain benchmarks at the finance and accounting boundary, spanning question answering, complex reasoning, **audit workflows**, spreadsheet-centric processes, and agentic research tasks. FinanceQA (Mateega et al., 2025) proposes a benchmark to evaluate LLM financial analysis capabilities, representing the broader class of finance-domain benchmarks. XFINBENCH (Z. Zhang et al., 2025) targets complex financial problem solving and reasoning, emphasizing multi-step reasoning structures and evaluation protocols suitable for harder, compositional tasks. FinMaster (Jiang et al., 2025) moves toward full-pipeline financial workflows, i.e. more end-to-end tasks rather than isolated questions, and therefore motivates evaluation stacks that can assess intermediate artifacts and process quality, not just final answers (see also (J. Zhou et al., 2026)). Finch (Dong et al., 2026) benchmarks finance and accounting across spreadsheet-centric enterprise workflows, which is highly salient for accounting settings where tasks are naturally expressed through tabular structures and spreadsheet operations. AuditBench (R. Wang et al., 2025) focuses specifically on LLM capability in financial statement auditing, aligning tasks and evaluation to auditing settings and thus illustrating how domain benchmarks can reflect professional practice constraints and documentation standards. The Finance Agent Benchmark (Bigeard et al., 2025) benchmarks LLMs on real-world financial research tasks, reflecting a trend toward agentic evaluation where models must plan, retrieve information, and assemble structured outputs. This is relevant to accounting benchmarking insofar as modern accounting work frequently involves multi-step workflows and tool use (document search, calculation, policy lookup, journal entries). Rosero et al. (2025) evaluate financial literacy using domain-specific languages for plain text accounting, illustrating an alternative path. Constraining model outputs into structured or semi-structured formats to improve evaluability and reduce ambiguity in grading. Across these benchmarks, the key differentiators include: **task coverage** (QA vs workflows), artifact modality (pure text vs spreadsheet/structured outputs), evaluation logic (scmc-formula and LLM-as-judge), and intended users (research vs enterprise operations). In parallel to benchmark construction, **a second literature stream evaluates LLM** ca-

pability using professional certification exams and educational accounting cases, often focusing on what performance implies for automation risk, education, and professional practice.

Bommarito et al. (2023) evaluate GPT models against CPA-related assessments and report results that differentiate performance by skill type, emphasizing that apparent competence may be **domain- and skill-specific rather than uniform**. Eulerich et al. (2024) extend this approach to multiple certifications (CPA, CMA, CIA) and show that performance changes substantially **across model versions and prompting/tooling regimes** (GPT-4 vs GPT-3.5, few-shot training, tool use). Their evidence is particularly important for benchmarking because it illustrates that scores are not stable properties of a model alone, they are functions of the **full system configuration**. In accounting education contexts, Cheng et al. (2024) investigate ChatGPT performance on educational accounting cases and discuss capabilities and limitations with implications for pedagogy and assessment design. Such studies highlight that educational case performance can diverge **from real-world accounting work products**, but they provide relevant tasks for reasoning, explanation, and judgment. Gupta et al. (2025) analyze LLM performance on Chartered Accountancy-related assessments, reinforcing the broader point that performance depends on local exam designs, content distributions, and evaluation formats. Together, the exam- and education-based literature suggests two implications for accounting benchmarking. First, benchmarks should distinguish knowledge recall from judgment under uncertainty and artifact correctness (correct journal-entry recognition and measurement). Second, evaluation should **explicitly track confidence**, consistency, and error patterns, because professional risk often arises **from plausible but wrong outputs** rather than from obvious failure.

The above literature reveals a clear gap that an accounting-focused benchmarking framework can fill if it is designed around accounting-specific constructs and research needs. Accounting work often culminates in normatively constrained artifacts that must satisfy standards and consistency. Some benchmarks already move closer to practice settings (spreadsheets, auditing), but the overrepresentation of finance-specific benchmarks and the accounting construct still benefits from explicit definition and operationalization. Accordingly, this paper should position the contribution as a reproducible, modular accounting benchmarking framework that integrates tasks, supports controlled experimental manipulations, and reports not only accuracy but also reliability and error diagnostics.

In a study published in November 2025 evaluating **445 LLM benchmarks** (Bean et al., 2025), 29 expert reviewers analyzed the strengths and weaknesses and the steps necessary to set up and implement good benchmarks. In particular, standardization in the academic environment for the **continuous improvement of benchmarks is inherent**. The

value of the benchmark lies in answering the question of whether it is a good proxy for the real-world phenomenon it attempts to measure. This is called construct validity. The construct validity must be measured to determine whether the model is intelligent (enough) or not. What the authors have done can be described as benchmark benchmarking leading to eight recommended steps to follow when creating benchmarks. First, a clear definition of the phenomenon that is going to be measured should be provided. If the task has one or more sub-components, every sub-component needs to be measured separately (Bean et al., 2024). This leads to the requirement of only measuring the task itself. Unrelated tasks or constraints that may affect the results of the benchmark should be neglected. Building a representative dataset is very important for accurately testing the LLM. This also includes incorporating permutations or variants within the data to test the LLM’s sensitivity, as LLMs can be very sensitive to small variations in prompts (Mizrahi et al., 2024; Zhuo et al., 2024). This also includes data contamination, which could be the case when reusing data and should be documented and checked during benchmarking (K. Zhou et al., 2023). Finally, the results of the benchmarking should also be evaluated using statistical methods and an error analysis should be performed to complete the process. This allows us to determine which and how the LLM mainly makes mistakes and how these can be corrected.

3 Accounting Benchmark Architecture and Execution Pipeline

AccountingBench defines the core phenomenon as accounting problem-solving competence in large language models. This competence includes the model’s ability to (i) understand structured and unstructured accounting and accounting related questions, (ii) apply normative accounting standards such as local GAAP, IFRS, as well as local tax law, (iii) perform reasoning and quantitative analysis, and (iv) produce logically consistent and interpretable responses. This operationalization ensures that the measured task corresponds to a professional, rule-based cognitive task like that faced by human accountants and accounting students. The operational definition is implemented through task templates and the evaluation. Single- and multiple-choice tasks measure factual knowledge and rule recognition, while open-ended tasks measure interpretative reasoning, justification, and analytical explanation.

Accounting reasoning is a construct consisting of factual retrieval, normative rule application, and explanatory reasoning. The framework measures each component separately through different task types: factual correctness via multiple-choice scoring

(Penalized Accuracy score), logical coherence and content alignment via open-ended evaluation (LLM-as-a-judge), and calibration of model confidence through self-assessed confidence score outputs. LLM-as-a-judge is used as a qualitative evaluator, consistent with emerging benchmarking practice, and we mitigate circularity by using structured prompts and confidence calibration. Benchmark prompts are designed to minimize the activation of unrelated cognitive abilities such as general world knowledge, common-sense reasoning, or linguistic style imitation. All questions are domain-specific and avoid generic reasoning keywords that could advantage models trained extensively on non-accounting corpora. In addition, the benchmark prompts templates standardize phrasing and structure to avoid syntactic variance that could confuse results. The conceptualization of accounting reasoning builds on research in accounting judgment and decision-making, which characterizes expert performance through rule-based interpretation, analytical structuring, and evidence-based justification. These components directly inform the design of the benchmark’s task types and scoring procedures (Libby & Luft, 1993; Nelson & Tan, 2005).

To follow the proposed structure for introducing new benchmarks, we conceptualized the accounting LLM benchmark ‘AccountingBench’. The pipeline consists of seven layers (see Figure 1):

1. Dataset definition and metadata stored in a ground-truth spreadsheet;
2. Model-input preparation via an export script that creates a model-call template while preserving the spreadsheet structure but not consisting the ground truth answers;
3. A three-trial inference layer producing structured JSON outputs (answer and self-reported confidence of the LLM);
4. A consolidation layer that aggregates trials (majority vote for single-/multiple-choice, averaging with tolerance checks for numeric outputs, and a consolidation call for open-text drafts);
5. An evaluation layer combining scoring for choice tasks, tolerance-based numeric comparison, and rubric-guided LLM-as-a-judge grading for open outputs;
6. Excel-safe logging to append results without overwriting templates; and
7. A run registry and output repository enabling auditability, replication, and stability analysis across models and configurations.

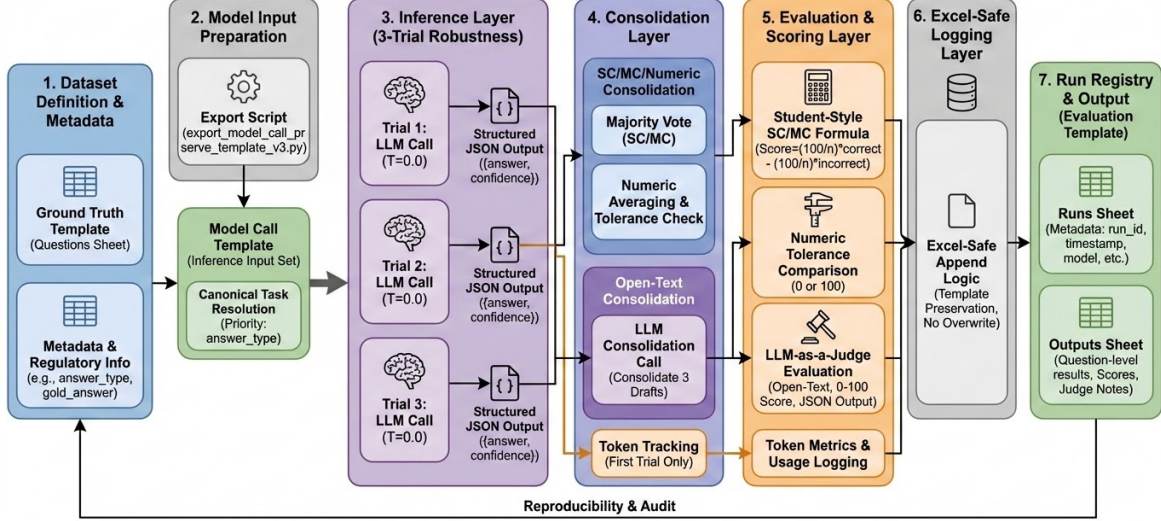


Figure 1: Accounting LLM benchmarking system architecture

We implement the benchmark as a template-driven system that separates dataset specification, model execution, and evaluation and logging. This separation ensures that task definitions and gold labels remain stable while enabling controlled variation of model configurations. The benchmark begins with a ground-truth spreadsheet that stores each item’s question, gold answer, and required metadata (e.g., task type, answer format, regulatory context). An export script generates a model-call template from the ground-truth sheet while preserving the spreadsheet structure, thereby enforcing consistent inputs across runs and models. Each task is executed in three independent trials. Each of these trials produces a structured JSON output containing the predicted answer and a self-reported confidence score. Trial outputs are consolidated using task-appropriate logic: majority vote for single- and multiple-choice tasks, numeric averaging with tolerance checks for quantitative tasks, and draft consolidation for open-text responses. The evaluation layer then applies scoring for choice tasks that penalizes overinclusive answering and LLM-as-a-judge grading for open outputs, producing a 0-100 score per item. We implement GPT-5 mini as a static judge for all runs and tasks to ensure comparability across evaluated models. Open-text responses are rated on a 0–100 scale based on correctness and completeness following task-specific grading criteria. All runs are appended via an logging mechanism that preserves templates and creates a run registry. The resulting output repository contains run-level metadata (model, timestamp, configuration) and item-level results (scores, confidence, judge notes), enabling replication and robustness analyses. This instrumentation is central to our study goal of mapping accounting tasks into automation-ready versus augmentation-only categories based on both performance and stability.

We apply a student scoring mechanism that rewards correct selections and penalizes

incorrect selections to discourage overinclusive answering in single- and multiple-choice tasks. This is particularly important in accounting settings where plausible but wrong treatments can create false assurance. The result is calculated as follows:

$$\text{SC/MC-Score} = \frac{100}{c} \cdot n_c - \frac{100}{c} \cdot n_i \quad (1)$$

where c denotes the number of correct alternatives, n_c the number of correct marks by the model, and n_i the number of incorrect marks by the model.

Numeric tasks are scored via tolerance-based comparison, assigning full credit when the consolidated numeric answer falls within a predefined tolerance band around the **gold value** and zero otherwise (e.g. ± 1.00). For journal entries, the LLM-as-a-judge protocol additionally evaluates **artifact validity** (e.g. debit-credit consistency and appropriate account selection). This approach is designed for scalability, but we explicitly treat judge scoring as an evaluative component that requires transparency (disclosure, prompt disclosure) to mitigate evaluator bias. We do acknowledge that LLM-as-a-judge evaluations do have their limitations due to AI hallucinations and different biases (inherent bias, length bias) as well as lack of domain expertise (Lee et al., 2026; T. Zhang et al., 2020; Zhao et al., 2019). The reason we nevertheless decided on an LLM-as-a-judge evaluation at this stage of the benchmark is the technical simplicity of implementation within the AccountingBench architecture and the rapid generation of evaluations compared to the manual evaluation of LLM outputs by domain experts. This saves scarce time resources and prevents the goal of automation from being halted again by **human intervention**.

All experiments create a run registry capturing run-level metadata (e.g., run ID, timestamp, model configuration), as well as an outputs sheet capturing item-level outputs (raw and consolidated), scores, judge notes, confidence values, and token/usage metrics where available (Figure 1, Layer 7). This design supports the replication of published results, post-hoc analysis of error patterns and stability. Taken together, the dataset, inference, consolidation, and scoring/logging layers implement a benchmark pipeline that is not only a model comparison tool, but an instrumented research platform for identifying which accounting tasks are candidates for automation and which remain augmentation-only due to error modes or instability.

The dataset for the benchmark draws from **accounting sources**, including secondary school and university-level textbooks, local tax and audit institutions professional exams and practical training materials used **in CPA preparation**. This multi-source sampling strategy ensures representativeness across the spectrum of difficulty, and problem type. The stratification follows recommendations from benchmark design research, which emphasizes that task difficulty must reflect the underlying construct and

be sufficiently heterogeneous to detect performance differences across models (Bean et al., 2025; Reuel et al., 2024). Each sample undergoes **expert review** by accounting academics to verify conceptual correctness, regulatory compliance, and clarity. Human verification is a design element widely regarded as essential for maintaining content validity in domain-specific benchmarks, particularly in high-stakes fields such as medicine, law, and finance, where automated or crowd-sourced labeling has been shown to produce conceptual drift and inconsistent ground truth (Guha et al., 2024; Q. Jin et al., 2019). LLMs are known to exhibit sensitivity to structural differentiations, **including variations in phrasing, option order**, or syntactic structure. Benchmarking literature consistently identifies such structural sensitivity as a source of construct-irrelevant variance and recommends introducing controlled item variants to test robustness and prevent models from exploiting superficial cues (Bommasani et al., 2023; Mizrahi et al., 2024; Patwardhan et al., 2025; Zhuo et al., 2024). Adapted exam questions provide realism and difficulty alignment but may lack explanatory depth. Conversely, textbook-based questions enable controlled reasoning but may **introduce stylistic bias**. Benchmarking studies note that such trade-offs are unavoidable in early-stage domain benchmarks and explicitly recommend combining multiple source types to balance ecological validity with methodological control (Eriksson et al., 2025; Gevers et al., 2025). By integrating both exam-based and textbook-based items, the accounting benchmark mirrors these design principles and achieves a measured equilibrium between realism and controlled construct measurement. The scope of the benchmark covers three major accounting subdomains: financial accounting, managerial accounting, and taxation.

Model contamination, which is prior exposure to benchmark items during pretraining, poses a major validity risk. AccountingBench has to maintain a sealed subset of test items not publicly released to preserve uncontaminated evaluation for future models. This aligns with best practices from HELM and GDPval (Bommasani et al., 2023; Patwardhan et al., 2025), ensuring the benchmark retains value over time. A domain specific error taxonomy is under development, distinguishing factual errors (misstated rules), reasoning errors (invalid logic), and numerical miscalculations. Such analysis provides insights into why a model fails, enhancing the explanatory value of benchmark results. Planned regression analysis is going to examine whether performance varies with question length or difficulty labels throughout different models. This identifies whether performance differences are due to **domain understanding** or **general linguistic complexities**.

After publication of the dataset and code, internal and external feedback will be addressed, resulting in minor and major changes to the benchmark; for major changes in the dataset or code, **new versions will be published**. It is important to document processes and ensure easy access to the data and reproducibility. In the final phase

of a benchmark, a “retirement plan” (Reuel et al., 2024) can be communicated and the data and documentation archived. Since the goal is to develop a continuously evolving evaluation instrument rather than a one-time snapshot, a publicly accessible leaderboard enables ongoing comparison of models across different LLM families and provider ecosystems. The leaderboard will be extended as additional models are evaluated over time, and the dataset continuously adapted.

4 Data

Our benchmark dataset comprises 520 validated accounting tasks, maintained through a structured annotation workflow implemented in an Excel-based ground-truth template. Each benchmark instance is represented as a single row in the sheet and follows a consistent schema with columns grouped into: task specification (`task_type`, `task_format`, `prompt`, optional `context`, optional `attached_files`), response space definition (optional `options` for choice-based formats), **gold supervision** (`answer_type`, `gold_answer`, optional `acceptable_variants`, optional `numeric_tolerance`), and evaluation guidance (`grading_criteria`). In addition, each item is richly annotated with domain and provenance metadata: `regulatory_framework` and `applicable_regulatory_year` (capturing the normative reference system), `category` and `skills` (competency tags), `education_level`, and `source` of the task. Finally, the template enforces traceability and auditability through `created_by`, `validated_by`, `validation_status`, and `last_updated`. Each task has its own unique key `question_id`.

Topical coverage is organized into three accounting super-categories: tax (349 / 67.1%), financial accounting (122 / 23.5%), and management accounting (49 / 9.4%). Task formats span both selected-response and generative evaluations: open-text responses constitute the largest share (300 / 57.7%), followed by multiple-choice (155 / 29.8%), single-choice (41 / 7.9%), journal-entry (16 / 3.1%), and open numeric (8 / 1.5%). This mix supports benchmarking along complementary axes: discrete option selection under distractors, and free-form reasoning and explanation where correctness depends on semantic adequacy and, where applicable, **structured accounting outputs**.

The majority of instances target interpretation-of-law reasoning (441 / 84.8%), reflecting the normative character of accounting and tax tasks. A secondary cluster focuses on calculation tasks (54 / 10.4%), capturing quantitative transformations such as recognition and tax-base derivations, where numeric rigor matters. A smaller subset comprises journal-entry tasks (25 / 4.8%), which require producing structured book-keeping outputs. Of these 25 journal-entry tasks, 16 use a structured journal-entry question format; the remaining 9 are presented as open-text questions and are there-

fore counted under “Open Text” in the question-format breakdown, explaining the discrepancy between the task-type count (25) and the question-format count (16) for this category.

Regarding educational provenance, the dataset draws primarily from professional examination materials (348 / 66.9%), supplemented by university master’s-level coursework (116 / 22.3%) and secondary vocational materials (56 / 10.8%). The regulatory scope covers five frameworks: Austrian tax law (388 / 74.6%), mixed accounting frameworks (62 / 11.9%), national GAAP (34 / 6.5%), IFRS (25 / 4.8%), and mixed accounting plus tax (11 / 2.1%).

Table 2 provides a full overview of the dataset composition.

Table 2: Dataset composition overview ($n = 520$ tasks)

Dimension	Category	Count	Share
Accounting domain	Tax	349	67.1%
	Financial Accounting	122	23.5%
	Management Accounting	49	9.4%
Question format	Open Text	300	57.7%
	Multiple-Choice	155	29.8%
	Single-Choice	41	7.9%
	Journal Entry *	16	3.1%
	Open Numeric	8	1.5%
Task type	Interpretation of Law	441	84.8%
	Calculation	54	10.4%
	Journal Entry *	25	4.8%
Education level	Professional Examinations	348	66.9%
	University Master’s	116	22.3%
	Secondary Vocational	56	10.8%
Regulatory framework	Austrian Tax Law	388	74.6%
	Mixed Accounting Framework	62	11.9%
	National GAAP	34	6.5%
	IFRS	25	4.8%
	Mixed: Accounting + Tax	11	2.1%

* The task-type category “Journal Entry” contains 25 tasks, while the question-format category contains only 16. The discrepancy arises because 9 of the 25 journal-entry tasks are presented in open-text format and are therefore counted under “Open Text” in the question-format breakdown.

5 Results

AccountingBench reports evaluation results across ten models from five providers—gpt-5.4, gpt-5.2, claude-opus-4-6, claude-sonnet-4-6, gpt-5-mini, Mistral-Large-3, grok-4-fast, gpt-4o, DeepSeek-V3.2-2, and mercury-2—using a harmonized inference and scoring pipeline applied to all 520 validated tasks. We begin with a cost comparison table to contextualize performance in terms of practical deployment considerations, motivating a cost–quality perspective when interpreting the results. We then present overall performance and stratified outcomes across key dataset slices.

Table 3: Cost comparison of evaluated models (prices as of March 2026). Average cost per task computed as $(\text{input tokens} \times p_{\text{in}} + \text{output tokens} \times p_{\text{out}})/10^6$, averaged over all 520 tasks.

Model	Organization	In \$/1M	Out \$/1M	Avg. Cost/Task
claude-opus-4-6	Anthropic	\$5.00	\$25.00	\$0.0282
gpt-5.4	OpenAI	\$2.50	\$15.00	\$0.0239
gpt-5.2	OpenAI	\$1.75	\$14.00	\$0.0179
claude-sonnet-4-6	Anthropic	\$3.00	\$15.00	\$0.0156
gpt-4o	OpenAI	\$2.50	\$10.00	\$0.0032
gpt-5-mini	OpenAI	\$0.25	\$ 2.00	\$0.0031
Mistral-Large-3	Mistral AI	\$0.50	\$ 1.50	\$0.0014
grok-4-fast	xAI	\$0.20	\$ 0.50	\$0.0012
mercury-2	Inception Labs	\$0.25	\$ 0.75	\$0.0009
DeepSeek-V3.2-2	DeepSeek	\$0.28	\$ 0.42	\$0.0003

Table 4: Main results: overall and category-level mean final score (% , $n = 520$ tasks). [†]Scored on 465/520 tasks (55 tasks excluded: no parseable `final_answer`). [‡]Scored on 516/520 tasks (4 excluded). [§]Scored on 519/520 tasks (1 excluded).

Rank	Model	Overall	Fin. Acc.	Mgmt. Acc.	Tax
1	gpt-5.4	72.2	78.1	88.1	68.0
2	gpt-5.2	68.6	77.3	87.8	62.9
3	claude-opus-4-6 [†]	63.3	74.4	88.5	57.1
4	claude-sonnet-4-6	63.1	77.6	88.3	54.4
5	gpt-5-mini	58.3	70.5	82.7	50.6
6	Mistral-Large-3 [‡]	52.7	69.8	85.5	42.0
7	grok-4-fast	49.1	64.3	84.5	38.8
8	gpt-4o	48.9	67.1	79.8	38.2
9	DeepSeek-V3.2-2 [§]	44.8	58.9	74.1	35.8
10	mercury-2	43.8	63.1	77.2	32.4
<i>Average (all models)</i>		56.5	70.1	83.7	48.0

Table 4 reports overall benchmark performance as the mean final score across all

$n = 520$ tasks. gpt-5.4 achieves the highest average score (72.2%), followed by gpt-5.2 (68.6%), claude-opus-4-6 (63.3%), claude-sonnet-4-6 (63.1%), gpt-5-mini (58.3%), Mistral-Large-3 (52.7%), grok-4-fast (49.1%), gpt-4o (48.9%), DeepSeek-V3.2-2 (44.8%), and mercury-2 (43.8%). The cross-model average score is 56.5%.

The mean benchmark performance stratified by topical category shows that the largest inter-model spread is observed in tax, where gpt-5.4 achieves 68.0%, compared to 62.9% (gpt-5.2), 57.1% (claude-opus-4-6), 54.4% (claude-sonnet-4-6), 50.6% (gpt-5-mini), 42.0% (Mistral-Large-3), 38.8% (grok-4-fast), 38.2% (gpt-4o), 35.8% (DeepSeek-V3.2-2), and 32.4% (mercury-2). In financial accounting, gpt-5.4 leads with 78.1%, followed closely by claude-sonnet-4-6 (77.6%) and gpt-5.2 (77.3%), while DeepSeek-V3.2-2 occupies the lower end (58.9%). Management accounting yields consistently high scores across all models (range: 74.1%–88.5%), indicating comparatively lower task difficulty and reduced between-model spread for this subdomain.

Stratified by regulatory framework, IFRS tasks approach ceiling for virtually all models (95.7%–100.0%), confirming that this framework is well represented in model pretraining. By contrast, Austrian tax law tasks continue to produce the largest model-to-model variation, highlighting this as the most discriminating task slice for future benchmark iterations. The complete results matrix across all 19 benchmark slices—spanning category, task type, answer type, regulatory framework, and education level—is reported in Table 5 (Appendix 8).

Regarding token usage and deployment cost (Table 3), models differ markedly in average response length. Average output tokens per task range from 681 (gpt-4o) to 1,997 (gpt-5.4). The highest-performing models are also among the most expensive: claude-opus-4-6 (\$0.0282/task), gpt-5.4 (\$0.0239/task), and gpt-5.2 (\$0.0179/task). Cost-efficient alternatives such as DeepSeek-V3.2-2 (\$0.0003/task), mercury-2 (\$0.0009/task), and grok-4-fast (\$0.0012/task) deliver lower absolute scores but at substantially reduced cost per task. Figure 2 visualises this cost–quality trade-off across all ten models.

Figure 3 reports confidence–performance calibration curves for five selected models representing the range of calibration patterns observed across all ten evaluated models. Across all models, a monotonically increasing relationship between reported confidence and mean score is observed: tasks answered at the highest confidence bin (0.95) yield mean scores ranging from 47.6% (mercury-2) to 100.0% (gpt-5.2). The alignment between confidence and accuracy is strongest for gpt-5.4 and gpt-5.2, which exhibit broad confidence distributions and strong score separation across bins. By contrast, gpt-4o and mercury-2 concentrate the vast majority of responses in the two highest confidence bins (0.85 and 0.95) despite substantially lower overall accuracy, indicat-

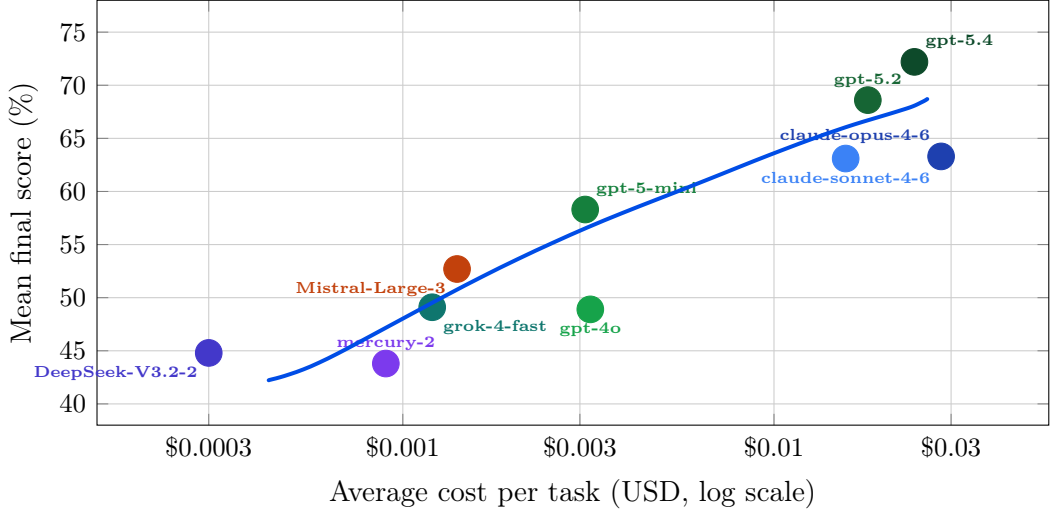


Figure 2: Mean final score vs. average API cost per task (log scale, $n = 520$ tasks). Each point represents one evaluated model. Cost computed as $(\text{input tokens} \times p_{\text{in}} + \text{output tokens} \times p_{\text{out}})/10^6$, averaged over all tasks (prices as of March 2026). Models towards the upper-left offer the most favourable cost–quality trade-off.

ing overconfidence patterns that are particularly consequential in accounting contexts where confident but incorrect outputs can create false assurance. Notably, gpt-5-mini exhibits the broadest confidence distribution among all evaluated models, spanning ten bins from 0.15 to 0.95, which is indicative of more **differentiated self-assessment across task difficulty levels**.

6 Limitations

The current version of AccountingBench comprises 520 validated tasks across three accounting domains and five regulatory frameworks. While this represents a substantial expansion, the dataset may produce elevated variance at the subcategory level; results for individual task slices should therefore be interpreted **with appropriate caution**. Cost constraints for repeated large-scale API evaluations remain a **practical consideration**, as documented throughout the pipeline design. These trade-offs are considered non-critical to construct validity because the underlying measurement principles—task representativeness and reliable scoring—remain intact. Future iterations will address remaining limitations through dataset expansion, fine-tuned evaluator models, and automated error classification.

Open-text responses are scored by an LLM judge (GPT-5 mini), **which introduces potential biases**; the judge may favor answers stylistically similar to its own outputs. A sample has been manually validated, but full human validation across **all 520 tasks**

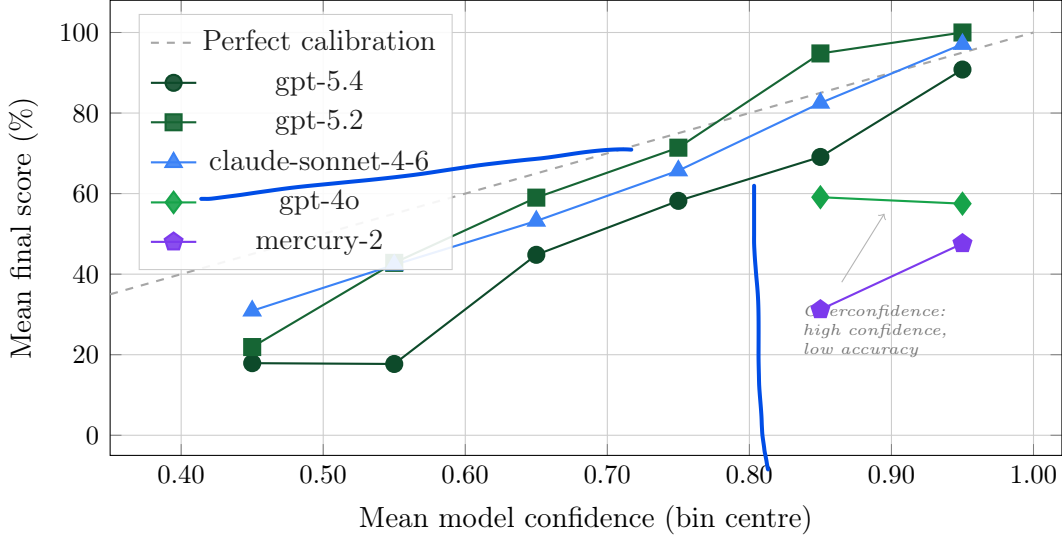


Figure 3: Confidence–performance calibration curves for five selected models ($n = 520$ tasks). Each point represents the mean final score for tasks within a confidence bin (bin width 0.1, plotted at bin centre); only bins with at least four observations are shown. The dashed diagonal indicates perfect calibration. Models gpt-4o and mercury-2 concentrate the vast majority of responses in the highest confidence bins (0.85 and 0.95) despite lower overall accuracy, indicating systematic overconfidence. Calibration curves for all ten models are provided in Figure 4 (Appendix 8).

has not been performed. More broadly, abstract constructs such as “accounting understanding,” “professional judgment,” or “regulatory interpretation” cannot be directly measured but must be inferred through narrow task formats and scoring procedures, which introduces an unavoidable gap between the construct of interest and its measurement (Bean et al., 2025). Without large-scale empirical validation through human performance distributions, the current design may capture only a subset of the cognitive processes underlying authentic accounting decision-making.

A further limitation relates to **output format sensitivity**: some models were not able to generate parseable answers **for a subset of tasks**. These tasks were automatically scored as zero but excluded from the calculation of the overall model score. Three models were affected: claude-opus-4-6 (**55 tasks excluded**, scored on 465/520), Mistral-Large-3 (**4 tasks excluded**, scored on 516/520), and DeepSeek-V3.2-2 (**1 task excluded**, scored on 519/520).

Another limitation relates to **contamination and model familiarity**. Given that many accounting standards, textbook explanations, and exam-style questions are widely available online, it is difficult to ascertain whether models have encountered similar or identical **content during training**. While the framework attempts to mitigate this risk through paraphrasing and structural variants, contamination cannot **be fully ruled out**,

and benchmark performance might partially reflect memorization rather than genuine reasoning (K. Zhou et al., 2023). This issue is particularly relevant in accounting, where authoritative phrasing from IFRS and local GAAP is stable and frequently reproduced across educational material, increasing the probability that LLMs have internalized these patterns during pretraining. AccountingBench will maintain a sealed subset of test items not publicly released to preserve uncontaminated evaluation for future model versions.

The framework is constrained by its static nature: it evaluates models under fixed prompts, fixed answer keys, and controlled difficulty tiers. However, accounting reasoning is dynamic and context-sensitive, often requiring iterative inquiry, the integration of additional evidence, and interaction with follow-up questions. Static benchmarks cannot capture this dialogic or exploratory dimension, which has become increasingly relevant as LLMs are deployed as interactive assistants rather than as one-shot answer generators. Results moreover reflect a single evaluation run in March 2026; model capabilities change over time and results should be re-evaluated periodically, especially following major model version releases.

The current evaluation covers ten models from five providers (OpenAI, Anthropic, Mistral AI, xAI, DeepSeek, and Inception Labs). While this represents a substantially broader cross-provider comparison than prior iterations, not all frontier models are yet included—in particular, additional open-weight models and proprietary systems remain to be evaluated. Reported performance differences should therefore be interpreted as a broad but not exhaustive snapshot of current model capabilities. Future work will extend model coverage and report updated cross-provider comparisons under harmonized inference settings to improve external validity and strengthen the generality of conclusions. The evolving nature of LLM architecture and frequent model updates by providers can additionally render benchmark results non-replicable over time, complicating longitudinal comparisons (Eriksson et al., 2025; Reuel et al., 2024).

7 Further Research

Future research will focus on expanding the benchmark into a fully dynamic operational evaluation environment and exploiting for empirical and methodological advancements in accounting research. The first priority is the systematic extension of the dataset across all subdomains. A publicly accessible leaderboard enables continuous comparison of models across different LLM families and provider ecosystems, and will be extended as additional models are evaluated over time. This interactive environment will show the results of the benchmark in a dynamic way to cluster the results

into varying categories as well as task formats to gain insightful knowledge on how models behave in these situations.

Subsequent research will also examine robustness by systematically varying prompt formulations, item structures, and linguistic features to quantify sensitivity effects across different LLM families. This includes testing multilingual variants and exploring jurisdictional extensions beyond IFRS, local GAAP, and local tax law. As the benchmark evolves, future studies may also evaluate the suitability of alternative scoring methods, finetuned evaluator models, and automated error taxonomies to increase the reliability and explanatory power of benchmark outputs. Beyond technical refinement, the benchmark offers a foundation for a new stream of empirical accounting research. Researchers can use it to analyze model behavior across specific standard-setting contexts, investigate LLM-assisted learning in accounting education, or study how regulatory complexity affects model reasoning. Finally, the benchmark will serve as a basis for longitudinal evaluations of model updates, enabling accounting researchers to monitor improvements, regressions, and stability over time. These avenues underscore the broader research potential of the benchmark and provide a roadmap for systematically building a domain-specific evaluation infrastructure for accounting.

8 Conclusion

This paper introduces AccountingBench, a structured benchmarking framework for evaluating LLMs on accounting problem-solving tasks under normative and artifact-based constraints. The framework combines a metadata-rich item bank, a template-driven multi-trial execution pipeline with structured outputs (including model-reported confidence), answer-type-specific consolidation, and a scoring system combining penalized choice scoring with rubric-guided LLM-as-a-judge evaluation for open outputs. The overall objective is to support auditability and replication and to enable an interpretable automation-versus-augmentation perspective rather than relying on aggregate accuracy alone.

Using 520 validated tasks, we provide a reproducible cross-provider baseline comparison of ten models under a harmonized setup. Results show systematic performance differences across models and substantial heterogeneity across task families and regulatory slices. Tax tasks and Austrian-law contexts remain the most discriminating slices, while management accounting yields near-ceiling performance across most models. The average model score of 56.5% and the top score of 72.2% (gpt-5.4) reinforce that practical adoption decisions in accounting should be grounded in task-class-

specific evidence rather than aggregate performance alone. Confidence calibration patterns further reveal that several models exhibit overconfidence at lower accuracy levels, which is particularly consequential in accounting settings where plausible but incorrect outputs can create false assurance.

Key limitations include the subcategory-level variance associated with the current dataset size, evaluator assumptions inherent in LLM-as-a-judge scoring, residual benchmark contamination risk, and the static nature of the evaluation design. Ongoing work will expand and version the item bank, introduce sealed test subsets and controlled prompt variants, extend model coverage, and refine the error taxonomy to further strengthen AccountingBench as a research instrument for assessing where LLMs can be used reliably for automation and where they should remain augmentation tools with human oversight.

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Appendix

Appendix A: Full Results Matrix

Table 5 reports mean final scores for all ten evaluated models across all benchmark slices: three category dimensions (tax, financial accounting, management accounting), three task-type dimensions (interpretation of law, calculation, journal entry),

four answer-type dimensions (multiple-choice, open text, single-choice, journal entry format), five regulatory framework dimensions (Austrian tax law, mixed accounting framework, national GAAP, IFRS, mixed accounting plus tax), and three education-level dimensions (professional examinations, master’s, secondary vocational school). Cells are colour-coded by score range: $\geq 90\%$, 75–90%, 60–75%, 50–60%, $< 50\%$.

Note on the Calculation column: The consistently low calculation scores across all models are largely driven by task composition rather than a general inability to perform calculations. Of the 54 calculation tasks, 45 are open-text, 8 are open-numeric, and 1 is multiple-choice. Performance is strong on the multiple-choice item ($\approx 95\%$) and moderate on open-numeric ($\approx 59\%$), but very low on open-text ($\approx 13\%$), which pulls down the category average. The open-text calculation tasks are drawn from a textbook and require extracting values from tabular content; models appear to struggle with reliably reading and structuring this tabular information.

Appendix B: Confidence–Performance Calibration: All Models

Figure 4 shows confidence–performance calibration curves for all ten evaluated models. Each subplot reports the mean final score per confidence bin (bin width 0.1, plotted at bin centre); only bins with at least four observations are included. The dashed diagonal indicates perfect calibration. Models are ordered by overall benchmark rank.

Table 5: Main results matrix: mean final score (%) across all benchmark slices ($n = 520$ tasks, March 2026). Colour coding: ≥90%
75–90% 60–75% 50–60% <50%. [†]Scored on 465/520 tasks. [‡]Scored on 516/520 tasks. [§]Scored on 519/520 tasks.

Model	Overall	Category			Task Type			Answer Type				Reg. Framework					Education		
		Tax	Fin. Acc.	Mgmt. Acc.	Interp. of Law	Calculation	Journal Entry	Multi-Choice	Open Text	Single-Choice	Journal Entry fmt.	Austrian Tax	Mixed Acc.	National GAAP	IFRS	Mixed Acc.+Tax	Prof. Exams	Master's	Sec. Voc.
gpt-5.4	72.2%	68.0%	78.1%	88.1%	77.8%	26.4%	72.4%	91.4%	60.7%	85.4%	74.1%	66.7%	88.2%	82.4%	98.0%	88.6%	68.3%	90.6%	58.3%
gpt-5.2	68.6%	62.9%	77.3%	87.8%	73.6%	24.9%	76.0%	91.1%	55.2%	80.5%	76.4%	62.0%	87.1%	83.1%	98.0%	86.4%	63.3%	90.0%	57.2%
claude-opus-4-6 [†]	63.3%	57.1%	74.4%	88.5%	67.2%	30.5%	74.3%	91.3%	52.7%	68.3%	77.5%	56.7%	89.3%	72.1%	100.0%	91.7%	57.8%	90.4%	56.5%
claude-sonnet-4-6	63.1%	54.4%	77.6%	88.3%	67.1%	26.1%	71.8%	89.6%	47.4%	73.2%	76.8%	54.6%	89.1%	80.5%	99.0%	78.8%	54.5%	90.7%	59.2%
gpt-5-mini	58.3%	50.6%	70.5%	82.7%	62.9%	19.5%	61.6%	85.6%	43.5%	65.9%	61.4%	50.4%	80.6%	75.0%	99.0%	67.4%	50.5%	85.7%	50.2%
Mistral-Large-3 [‡]	52.7%	42.0%	69.8%	85.5%	57.2%	20.7%	42.9%	88.3%	32.2%	70.7%	40.2%	42.1%	86.4%	70.4%	99.0%	74.2%	42.0%	89.2%	42.9%
grok-4-fast	49.1%	38.8%	64.3%	84.5%	51.9%	22.8%	57.5%	84.2%	28.1%	65.9%	63.9%	39.2%	79.7%	62.9%	97.3%	72.7%	39.1%	80.0%	47.5%
gpt-4o	48.9%	38.2%	67.1%	79.8%	53.9%	13.0%	38.5%	84.7%	28.8%	65.9%	41.1%	37.8%	80.8%	71.9%	100.0%	72.7%	37.5%	89.3%	36.1%
DeepSeek-V3.2-2 [§]	44.8%	35.8%	58.9%	74.1%	49.5%	12.4%	32.7%	80.1%	25.2%	65.0%	25.3%	35.4%	76.6%	53.5%	95.7%	56.1%	34.9%	81.2%	31.1%
mercury-2	43.8%	32.4%	63.1%	77.2%	47.3%	16.7%	40.1%	79.2%	22.4%	68.3%	46.1%	32.6%	77.9%	62.8%	99.0%	62.1%	31.9%	82.4%	37.9%

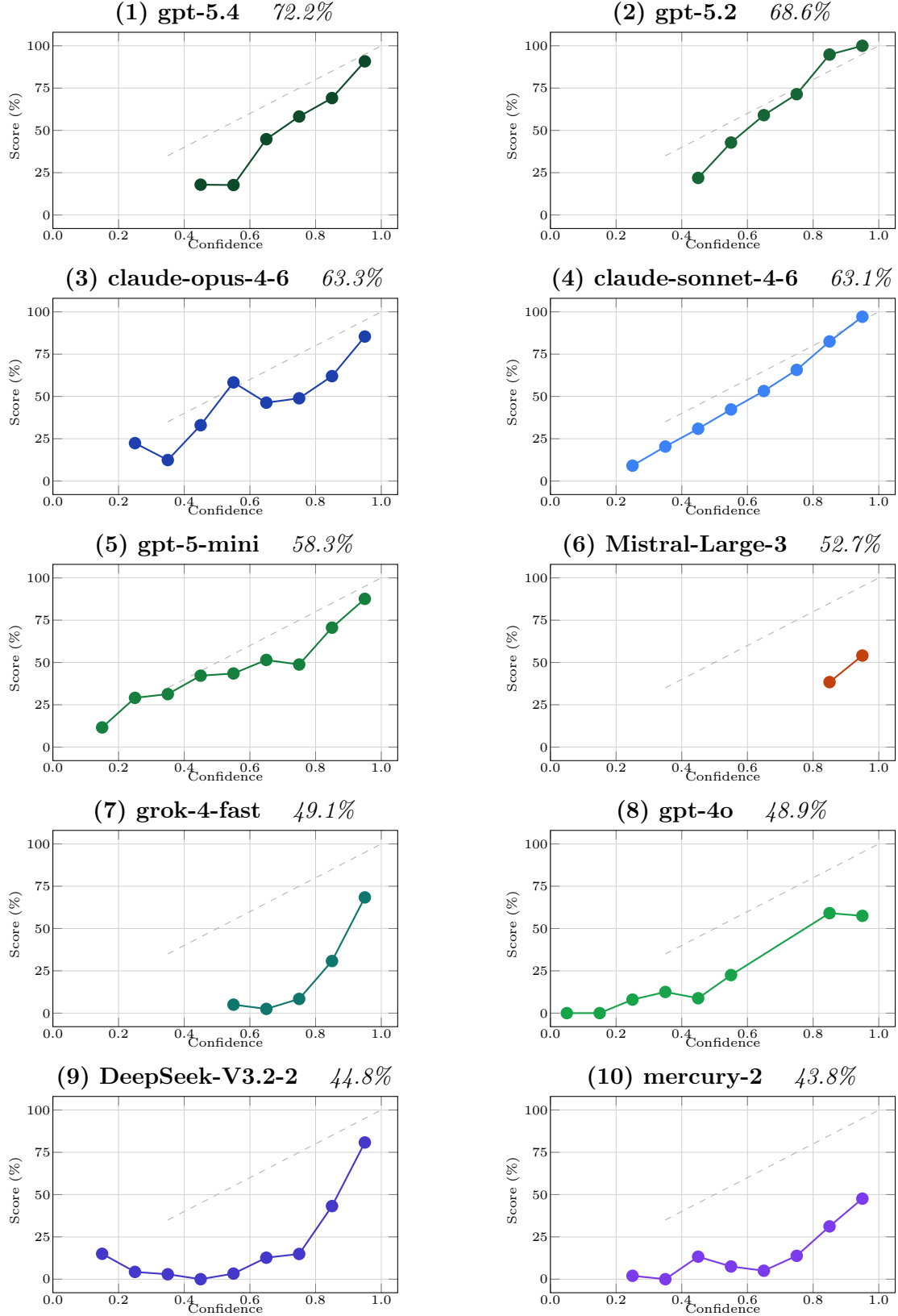


Figure 4: Confidence–performance calibration for all ten evaluated models, ordered by overall rank. Each point represents the mean final score for tasks in a confidence bin (bin width 0.1, plotted at bin centre); bins with fewer than four observations are omitted. The dashed diagonal indicates perfect calibration. Overall score shown in italics next to each model name.